

Filip Karika

JUNIOR SOFTWARE DEVELOPER | BACKEND & WEB DEVELOPMENT

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PROFESSIONAL SUMMARY

Junior Software Developer focused on backend and web development, currently studying at Unicorn University and 42 Prague. Through project work and technical instruction, I have gained experience across the development lifecycle — from solution design and architecture thinking to implementation, documentation, and teaching. Experienced in technical content creation and hands-on projects involving AI integration, Unix systems, networking, IoT, and full-stack development.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C, C++	Web: HTML, CSS, React, Node.js, Express.js, Electron, MongoDB, SQL
AI / Data: RAG, MCP, LLM integration	Tools / Platforms: Git, GitHub, Linux, Docker, Docker Compose, Node-RED, Arduino, MongoDB Atlas
Core CS: Algorithms, Data Structures, OOP	Languages: Slovak (Native), English (B2-C1)

SELECTED PROJECTS

AI Chatbot Integration for HARDWARIO Playground (*Electron, RAG, LLM*)

Bachelor's thesis focused on integrating an AI chatbot into the open-source HARDWARIO Playground application. The solution is designed to answer user questions based on company documentation and Node-RED-related knowledge using retrieval-augmented generation and LLM integration.

Transcendence (*Docker, React, Node.js, Express.js, MongoDB, RAG, LLM*)

AI-powered collaborative workspace inspired by Slack, focused on internal team communication, project organization, document handling, and AI-assisted retrieval over uploaded knowledge.

Minishell (*C*)

Unix project focused on building a custom shell inspired by Bash, with support for command execution, pipes, redirections, heredocs, built-in commands, environment variables, and signal handling. Designed with a modular architecture covering tokenization, parsing, and execution to reflect core Unix shell behavior.

SafeHome (*C, Node-RED, React, Node.js, Express.js, MongoDB, MongoDB Atlas*)

IoT-based home security system for detecting unauthorized entry using magnetic sensors, combining hardware communication, gateway logic, backend services, and frontend application features with real-time user notifications.

IRC Server (*C++98*)

Server project based on the IRC protocol, focused on implementing real-time client communication, channel management, command parsing, and server-side message routing in a multi-user environment.

RELEVANT EXPERIENCE

Web Development Instructor / Content Creator — Lionscz

July 2024 – July 2026

Create educational materials and manuals focused on web technologies and artificial intelligence, and teach students through structured lessons, technical explanations, and practical coding examples with the effective use of AI tools.

EDUCATION

Unicorn University, Prague - BSc in Software Development (2023 - Present)

42 Prague, Prague - Software Engineering Program (2024 - Present)

Charles University, Faculty of Mathematics and Physics, Prague - Informatics (2022 - 2023)

Súkromná stredná odborná škola (SSOSTA), Poprad - Digital Technologies (2018 - 2022)

SELECTED CERTIFICATIONS

Elements of AI — University of Helsinki (2025)	MongoDB Essentials — Unicorn University (2024)
Responsive Web Design — FreeCodeCamp (2024)	Business English B2-C1 — Unicorn University (2025)
Introduction to Software Architecture — Unicorn University (2024)	Introduction to Software Development — Unicorn University (2024)